

Derivatives and Fixed Income
PGE M1 Fall 2026

Jason Kang

Session 7: Option Basics



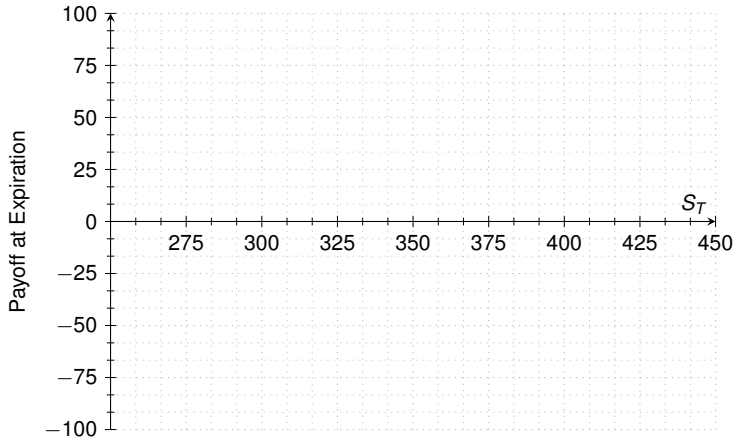
About me

- ▷ Office: B3.09
- ▷ Office Hours: E-mail for appointment or walk-in for short queries
- ▷ jason.kang@skema.edu

Recall: Long Forward



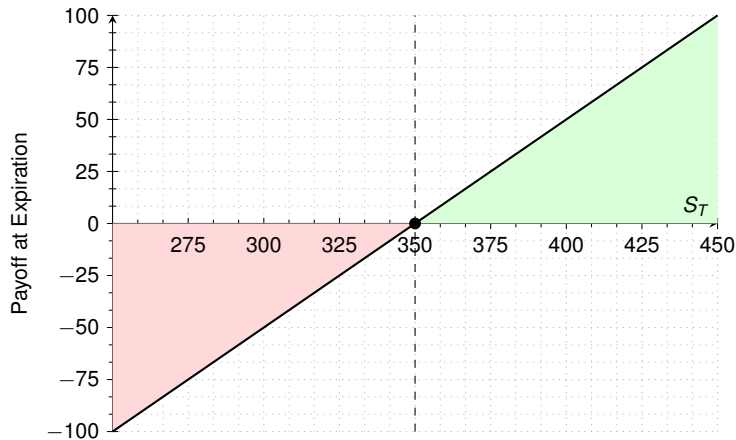
Suppose you entered a long forward position to **buy** Apple stock in one year at \$350. Draw the payoff diagram.



Recall: Long Forward



Suppose you entered a long forward position to **buy** Apple stock in one year at \$350. Draw the payoff diagram.



Call Option



A (European) call option is a contract that allows the holder/buyer of the call the the right, but not the obligation, to purchase the underlying asset at a pre-specified strike price K at maturity date T .

- ▷ Unlike a forward contract, you can walk away from the deal if you dislike the final price at T
- ▷ You agree on K and T today, and you “exercise” the option at the expiration date T .
- ▷ The owner of the call is said to be in a “long call” position.
- ▷ To enter the long call position, the holder **pays** an option premium.
- ▷ (later) American options allow the holder to exercise the option at **any** date before T , but we will focus on European options which can be exercised only at T .

Call Option



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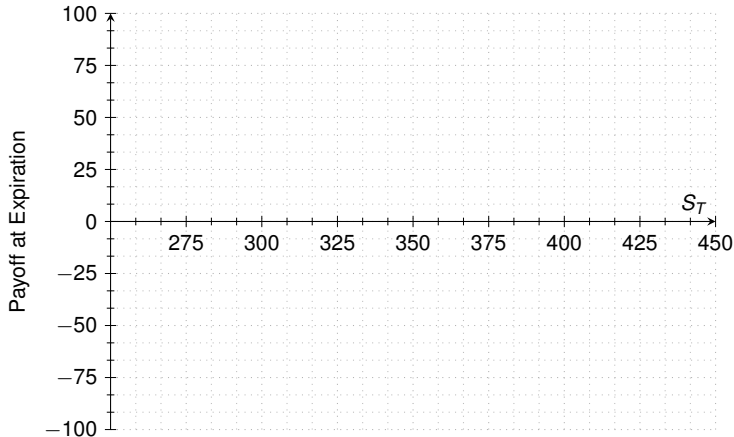
Example. Suppose you own a call option to purchase Apple stock in 1 year at price \$350. The price of entering into the contract was \$23.81.

- ▷ Underlying asset:
- ▷ Strike price: $K =$
- ▷ Maturity: $T =$
- ▷ Option premium: $c =$

Long Call



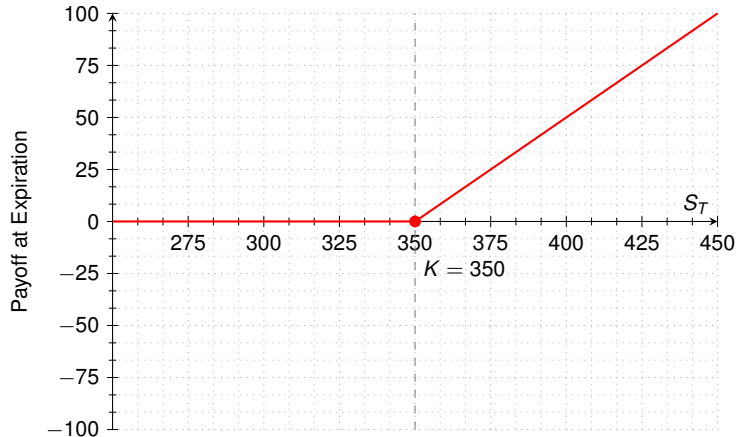
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Long Call



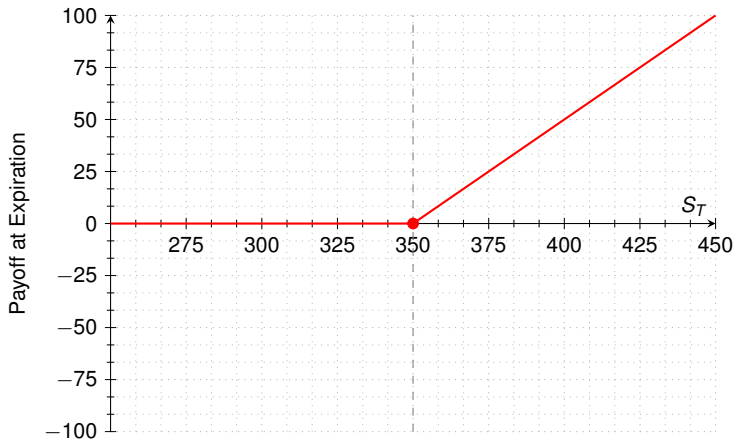
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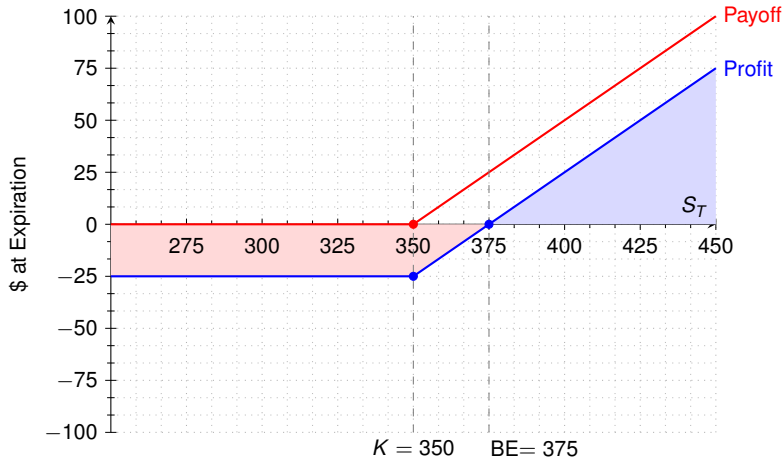
Suppose you entered a long call position to **buy** Apple stock in one year at \$350, paying a premium of \$23.81. Assume $r = 5\%$, so $23.81 \times 1.05 = 25$. Draw the payoff and profit diagrams.



Long Call



Suppose you entered a long call position to **buy** Apple stock in one year at \$350, paying a premium of \$23.81. Assume $r = 5\%$, so $23.81 \times 1.05 = 25$. Draw the payoff and profit diagrams.



Short Call



An option is a bilateral contract between two parties. If someone wants to buy the right to purchase an asset, someone else must be willing to sell the asset.

- ▷ When the holder of the call exercises the right to purchase the underlying, the seller is **obligated** to sell at K .
- ▷ The seller, or the writer of the contract, is said to be in a **short call** position.
- ▷ The seller **receives** an option premium

Short Call



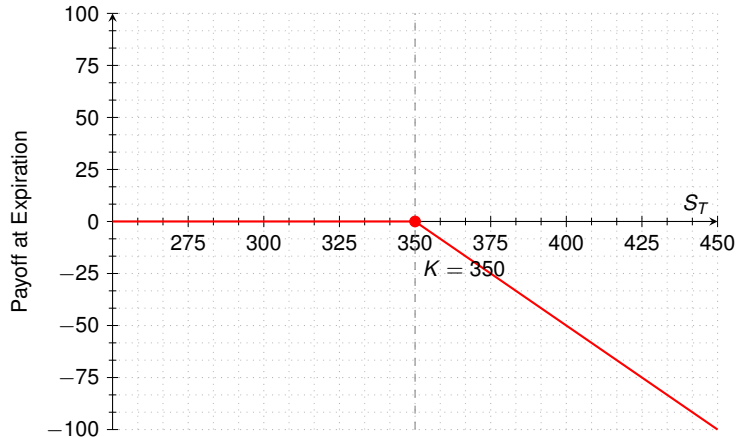
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Short Call



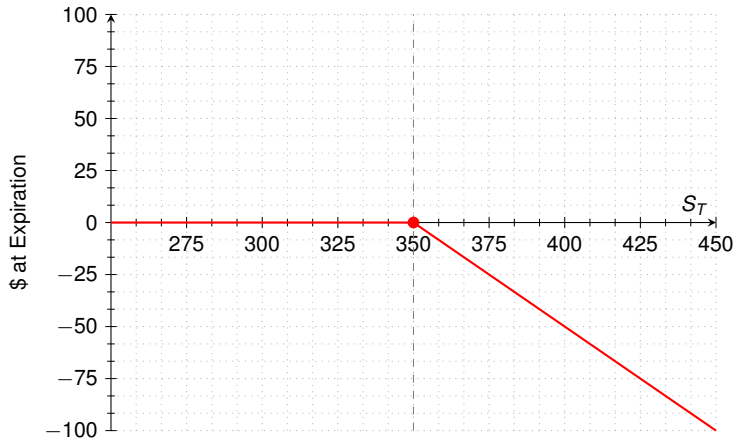
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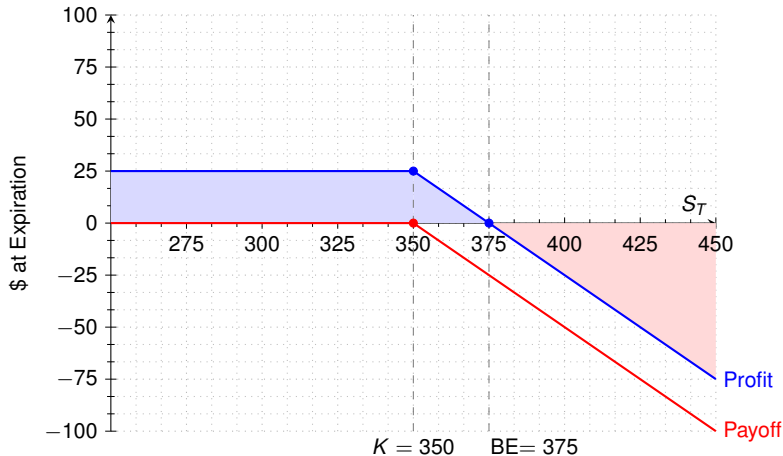
Suppose you entered a short call position to **sell** Apple stock in one year at \$350, receiving a premium of \$23.81. Draw the payoff and profit diagrams.



Short Call



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Who buys a call?

The option premium, or the price of the call contract, reflects the current “fair value” of the right to purchase at K at date T . Two people can enter an agreement due to

- ▷ Differences in subjective beliefs
- ▷ Differences in hedging needs
- ▷ Differences in risk tolerance/risk-aversion

Put Option



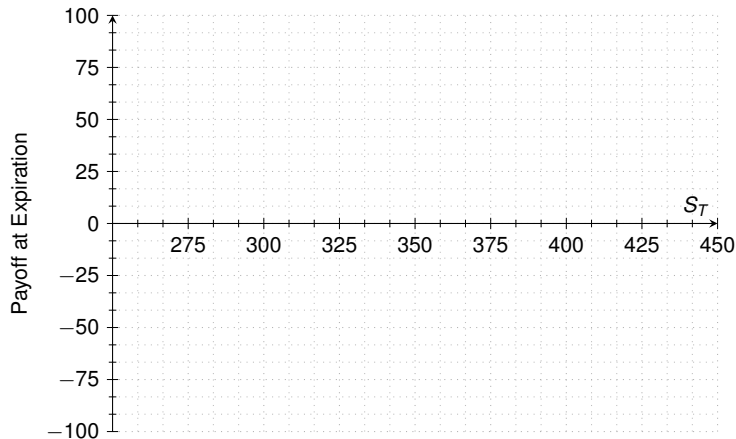
A (European) put option is a contract that allows the holder/buyer of the put the the right, but not the obligation, to sell the underlying asset at a pre-specified strike price K at maturity date T .

- ▷ Like before, you agree on K and T today, and you “exercise” the option at the expiration date T .
- ▷ The owner/buyer of the put is said to be in a “long put” position.
- ▷ To enter the long put position, the holder pays an option premium.

Long Put



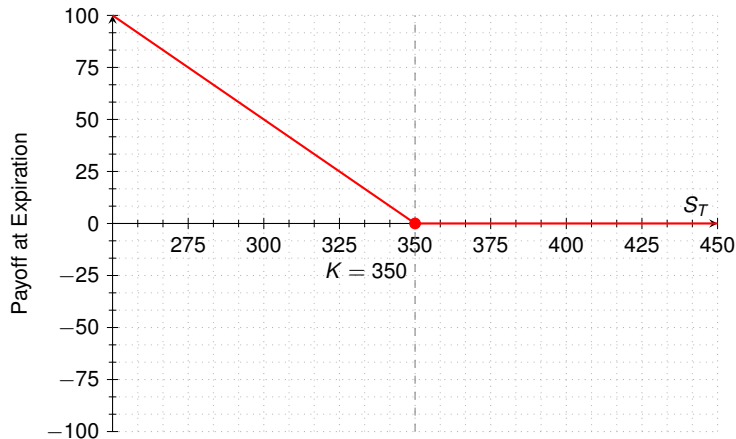
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Long Put



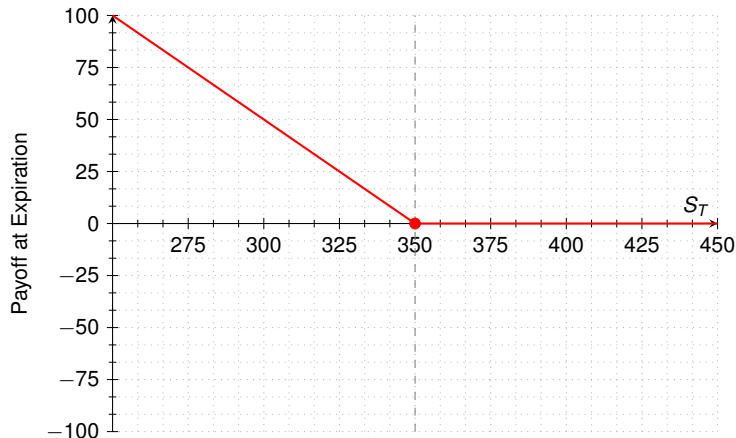
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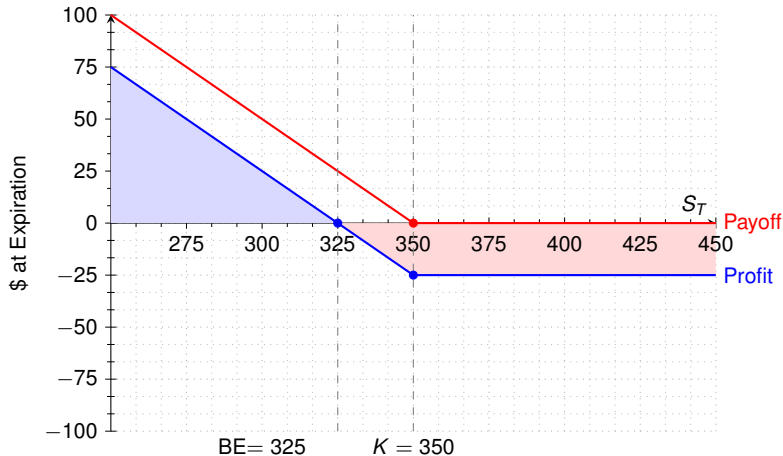
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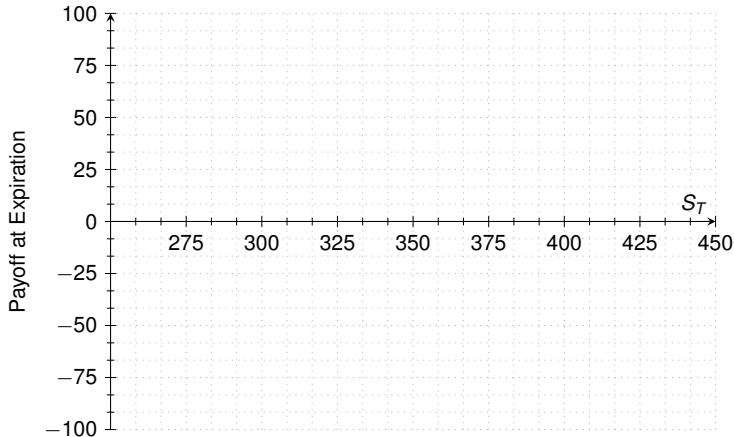
If the holder of the put exercises the right to sell at K , the seller/writer of the put is **obligated to buy at K** .

- ▷ In exchange, the writer of the put **receives** an option premium today
- ▷ The seller, or the writer of the contract, is said to be in a **short put** position.

Short Put



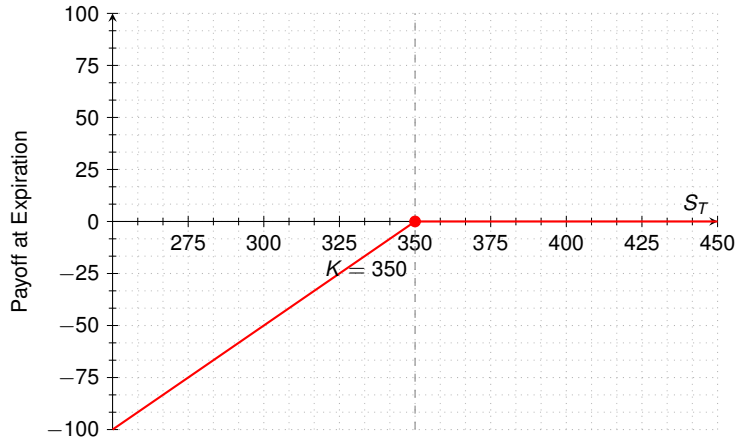
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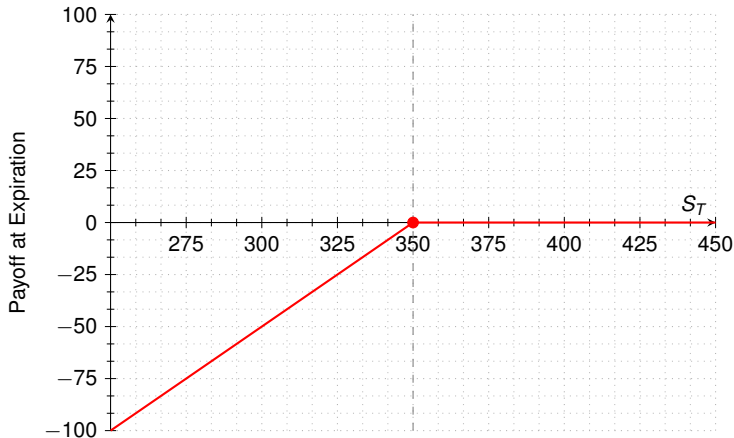
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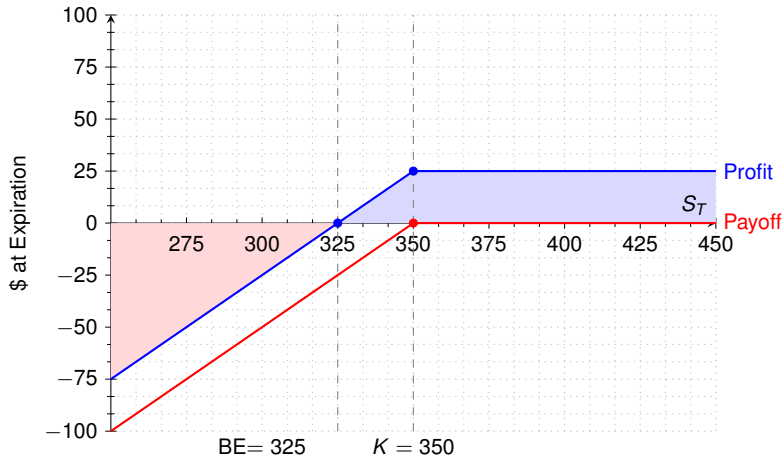
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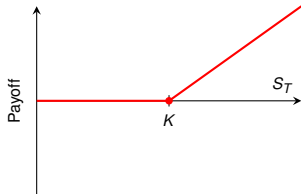
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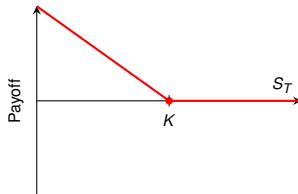
Payoff Diagrams: Calls and Puts



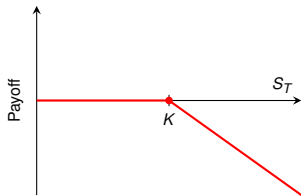
Long Call



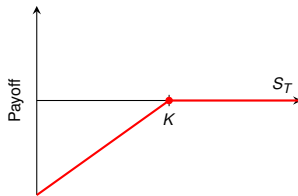
Long Put



Short Call



Short Put



Payoff Tables



	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Put	$-P$	$K - S_T$	0
Short Put	$+P$	$S_T - K$	0
Long Call	$-C$	0	$S_T - K$
Short Call	$+C$	0	$K - S_T$

Note, for a long call position, a payoff of $\max\{S_T - K, 0\}$ is equivalent to a payoff of $S_T - K$ when $S_T > K$ and 0 otherwise, as the option is not exercised.

Payoff Tables



	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Stock	$-S_0$	S_T	S_T
Short Stock	$+S_0$	$-S_T$	$-S_T$
Borrow (sell bond)	$+S_0$	$-S_0(1+r)^T$	$-S_0(1+r)^T$
Lend (buy bond)	$-S_0$	$S_0(1+r)^T$	$S_0(1+r)^T$
Borrow (sell bond)	$+\frac{K}{(1+r)^T}$	$-K$	$-K$
Lend (buy bond)	$-\frac{K}{(1+r)^T}$	K	K

Technically, you can borrow and lend any amount today but S_0 and $\frac{K}{1+r}$ are the ones relevant.